

WORK SHEET - 06

THERMOCHEMISTRY

Single correct

- Q.1 Calculate resonance energy of $\text{CO}_{2(g)}$ from the following data.
 $\Delta H_{\text{combustion}}$ of $\text{CO} = -280 \text{ kJ / mole}$ $\Delta H_f \text{CO} = -120 \text{ kJ/mol}$
 $\Delta H_{\text{atomisation}}$ of $\text{C}_{(\text{graphite})} = 720 \text{ kJ / mole}$ $\Delta H_{\text{BE}} \text{O} = \text{O} = 500 \text{ kJ mol}^{-1}$
 $\Delta H_{\text{BE}} \text{C} = \text{O} = 710 \text{ kJ /mol}$
 (A) 200 kJ (B) 220 kJ (C) 140 kJ (D) 100 kJ
- Q.2 The enthalpy of solution, of sodium and sodium oxide in large volume of water, are **-184** and **-238 kJ/mol**, respectively. If the enthalpy of formation of water is **-286 kJ/mol**, then what is the enthalpy of formation of sodium oxide? All the enthalpies are at 298 K and 1 bar pressure.
[Given : Reactions involved are]

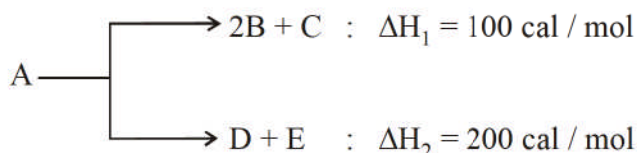
$$2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2(\text{g})$$

$$\text{Na}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} \quad]$$

 (A) +54 kJ/mol (B) -130 kJ/mol (C) -416 kJ/mol (D) +156 kJ/mol
- Q.3 Calculate the amount of heat released at constant pressure when 10 moles of carbon react with 6 moles of O_2 leaving none of the reactants.
 $\Delta H_{\text{combustion}} \text{C}_{(\text{graphite})} = -390 \text{ kJ/mole}$, $\Delta H_{\text{Combustion}} \text{CO}_{(\text{g})} = -250 \text{ kJ/mole}$
 (A) 1900 kJ (B) 750 kJ (C) 3900 kJ (D) 2450 kJ
- Q.4 Which of the following substances will have positive value of $\Delta H_{\text{formation}}$?
 (A) $\text{Br}_2(\text{l})$ (B) $\text{O}_3(\text{g})$ (C) ZnO(s) (D) $\text{H}^+(\text{aq})$
- Q.5 The bond enthalpies of C – C, C = C and C $\equiv$ C bonds are 348, 610 and 835 kJ/mol, respectively at 298K and 1 bar. The enthalpy of polymerisation per mole of 2-Butyne at 298K and 1 bar, as shown below, is

$$n \text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 (\text{g}) \longrightarrow -(\text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_2)_n (\text{g})$$

 (A) -123 kJ (B) -132 kJ (C) -139 kJ (D) -37 kJ
- Q.6 Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH. If enthalpy of neutralization of HCl by NaOH is -55.84 kJ/mol. Calculate $\Delta H_{\text{ionization}}$ of H_3PO_3 into its ions
 (A) 50.84 kJ/mol (B) 5 kJ/mol (C) 2.5 kJ/mol (D) None
- Q.7 For the reaction:



If initially 1 mol of A is taken & finally ratio of B & D is 8 : 1 what is total enthalpy change per mole of A reacted.

- (A) $\frac{1000}{9}$ cal (B) 180 cal (C) 1000 cal (D) 120 cal

- Q.8 When 100 ml 0.2 M KOH is mixed with 100 ml 0.2 M HCl, temperature of solution increases by $t_1^\circ\text{C}$ while when 300 ml 0.1 M KOH is mixed with 300 ml 0.1 M HCl then increase in temperature is $t_2^\circ\text{C}$ then which one is correct. (Assuming density as well as specific heat of final solutions are same)
 (A) $t_1 = t_2$ (B) $t_1 > t_2$ (C) $t_1 < t_2$ (D) Can't be predicted
- Q.9 Enthalpy for the reaction $\text{Ag}^+_{(\text{aq.})} + \text{Br}^-_{(\text{aq.})} \rightarrow \text{AgBr}_{(\text{s})}$ is -100 kJ . If magnitude of enthalpy of formation of $\text{Ag}^+_{(\text{aq.})}$ and $\text{Br}^-_{(\text{aq.})}$ is in the ratio of 5 : 6. While formation of $\text{Ag}^+_{(\text{aq.})}$ is an endothermic whereas formation of Br^- is an exothermic process. Enthalpy of formation of AgBr is -110 kJ/mole then the enthalpy of formation of $\text{Ag}^+_{(\text{aq.})}$ in kJ/mol .
 (A) 50 kJ (B) + 60 kJ (C) 100 kJ (D) 10 kJ
- Q.10 Two solids A and B having molar masses 200 & 300 react to form another solid C as shown.

$$2\text{A}_{(\text{s})} + 3\text{B}_{(\text{s})} \longrightarrow \text{C}_{(\text{s})}$$

 If $\Delta H^\circ_{\text{combustion}}$ of A, B and C are -200 , -300 & -400 kJ respectively then maximum heat which can be released at constant pressure if total 2600 gm of A and B is taken is given by
 (A) 200 kJ (B) 1800 kJ (C) 900 kJ (D) 450 kJ
- Q.11 When 4 gm graphite is burnt in limited supply of oxygen gas at 298 K and 1 bar, 9 Kcal heat is released and no solid residue is left. The gaseous product formed is not absorbed in aqueous KOH solution. When 6 gm graphite is burnt in excess of oxygen gas at 298 K and 1 bar, 48 Kcal heat is released and no solid residue is left. The enthalpy change of the following reaction at 298 K and 1 bar will be

$$2\text{CO}_2(\text{g}) \longrightarrow 2\text{CO}(\text{g}) + \text{O}_2(\text{g})$$

 (A) + 69 Kcal (B) + 39 Kcal (C) +138 Kcal (D) +57 Kcal
- Q.12 The standard enthalpy of atomisation of $\text{PCl}_3(\text{g})$ is 195 Kcal/mol. What will be the standard enthalpy of atomisation of $\text{PCl}_5(\text{g})$, if the bond dissociation energies of axial P-Cl bonds in $\text{PCl}_5(\text{g})$ are 10% lesser and the bond dissociation energies of equatorial P-Cl bonds in $\text{PCl}_5(\text{g})$ are 10% higher than the bond dissociation energies of P-Cl bonds in $\text{PCl}_3(\text{g})$.
 (A) 195 Kcal / mol (B) 325 Kcal / mol (C) 331.5 Kcal / mol (D) 318.5 Kcal / mol
- Q.13 Calculate change in enthalpy when 2 moles of liquid water at 1 bar and 100°C is converted into steam at 2 bar and 300°C . Assume H_2O vapours to behave ideally.
 [Latent heat of vaporisation of $\text{H}_2\text{O}(\text{l})$ at 1 bar and 100°C is 10.8 Kcal per mole]
 [R = 2 Cal / mol K]
 (A) 21.6 Kcal (B) 11.8 Kcal (C) 24.8 Kcal (D) 23.6 Kcal
- Q.14 One mole of magnesium in the vapour state absorbed 1300 kJ of energy. If the first and second ionization energies of Mg are 750 and 1450 kJ mol^{-1} respectively, the final composition of the mixture is
 (A) 59% Mg^+ and 41% Mg^{+2} (B) 41% Mg^+ and 59% Mg^{+2}
 (C) 62% Mg^+ and 38% Mg^{+2} (D) 38% Mg^+ and 62% Mg^{+2}

Comprehension :

Paragraph for question nos. 15 to 17

It is observed that when acid and bases react, some energy is released due to neutralization reaction. While in case of strong acids and bases, the energy released is greater, in case of weaker acids or bases, energy released will be lesser. The difference is attributed to energy required for ionization. From the above information and data below answer the questions that follow.

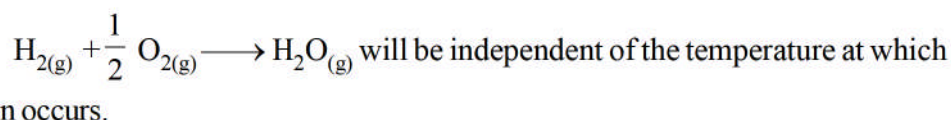
$\Delta H_{\text{neutralization SA/SB}} = -57.5 \text{ kJ/equivalent}$; $\Delta H_{\text{ionization}}$ of $\text{NH}_4\text{OH} = 10 \text{ kJ/mole}$;
 $\Delta H_{\text{ionization}}$ of $\text{H}_2\text{C}_2\text{O}_4 = 20 \text{ kJ/mole}$.



- Q.15 What will be the energy released when 500 ml of 0.1M H_2SO_4 solution reacts with 1L of 0.1M NH_4OH solution if the base is 30% ionized in the given solution?
 (A) 5.75 kJ (B) 4.75 kJ (C) 5.05 kJ (D) 5.45 kJ
- Q.16 What will be $\Delta H_{\text{neutralization}}$ of $\text{H}_2\text{C}_2\text{O}_4$ with NaOH per mole of $\text{H}_2\text{C}_2\text{O}_4$ if acid is assumed to be completely unionized in the solution?
 (A) 37.5 kJ/mole (B) -75 kJ/mole (C) -95 kJ/mole (D) -37.5 kJ/mole
- Q.17 What will be the final temperature attained if all the heat released in neutralization of 1L of 0.2M NH_4OH with 2L of 0.1M HCl increases the temperature of the final solution having density 0.95 gm/ml and specific heat capacity = $\frac{1}{3}$ J/gm°C if original temperature was 27°C. Assume weak base to be completely unionized.
 (A) 310°C (B) 300K (C) 310 K (D) 290 K

Assertion Reason

- Q.18 **Statement-1** : Assuming ideal gas behaviour of all gases involved, ΔH° for the reaction



Statement-2 : For an ideal gas, enthalpy of a fixed amount depends only on temperature.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

More than one correct :

- Q.19 Which of the following represent(s) $\Delta H_{\text{atomization, H}_2\text{O}(\text{s})}^\circ$?

- (A) $\Delta H_{\text{sublimation, H}_2\text{O}(\text{s})}^\circ + 2\Delta H_{\text{BE, O-H}}^\circ$ (B) $\Delta H_{\text{fusion, H}_2\text{O}(\text{s})}^\circ + \Delta H_{\text{vapourisation, H}_2\text{O}(\text{l})}^\circ$
 (C) $-\Delta H_{\text{formation, H}_2\text{O}(\text{s})}^\circ + \Delta H_{\text{BE, H-H}}^\circ + \frac{\Delta H_{\text{BE, O=O}}^\circ}{2}$ (D) $\Delta H_{\text{fusion, H}_2\text{O}(\text{s})}^\circ + \Delta H_{\text{atomization, H}_2\text{O}(\text{l})}^\circ$

- Q.20 In the thermite reaction used for obtaining energy, aluminium and ferric oxide react to form iron and aluminium oxide. From the given data identify the option(s) which is(are) **correct**:

[Given : $\Delta H_f^\circ \text{Al}_2\text{O}_3 = -390$ kJ/mol, $\Delta H_f^\circ \text{Fe}_2\text{O}_3 = -176$ kJ/mol,

Density of aluminium = 2.7 gm/ml, Density of $\text{Fe}_2\text{O}_3 = 3.2$ gm/ml]

(A) Maximum calorific value of the fuel can be 1000 J/gm.

(B) Maximum calorific value of the fuel can be $\frac{21.4}{7}$ kJ/ml

(C) 2.14 kg of the mixture can produce 2500 kJ of heat

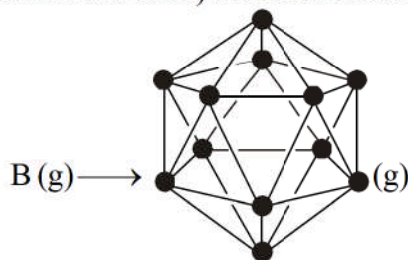
(D) 140 ml of the mixture can produce 280 kJ of heat

Subjective :

- Q.21 Calculate the enthalpy of ionisation of weak acid H_2A ($\text{H}_2\text{A} \rightarrow 2\text{H}^+ + \text{A}^{2-}$) in Kcal / mol, if enthalpies of neutralisation of HCl and H_2A by a strong base are 14 Kcal/eq and 11 Kcal/eq respectively.

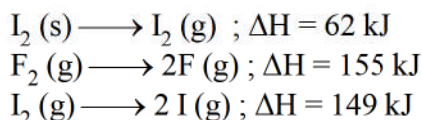


- Q.22 Boron exist in different allotropic forms. All allotropic form contains icosahedral units (icosahedral is a regular shape with 12 corners & 20 faces) with boron atoms at all 12 corners and all bonds are equivalent.



Calculate heat evolved at constant pressure (in kJ) per mole of boron atoms undergoing above change if $\Delta H_{BE}(B-B) = 200 \text{ kJ/mol}$.

- Q.23 The standard molar enthalpies of formation of $IF_3(g)$ and $IF_5(g)$ are -470 kJ and -847 kJ , respectively. Valence shell electron-pair repulsion theory predicts that $IF_5(g)$ is square pyramidal in shape in which all I-F bonds are equivalent while $IF_3(g)$ is T-shaped (based on trigonal-bipyramidal geometry) in which I-F bonds are of different lengths. It is observed that the axial I-F bonds in IF_3 are equivalent to the I-F bonds in IF_5 . Calculate the equatorial I-F bond strength (in kJ/mol) in IF_3 . Some other informations given are:



- Q.24 The standard molar enthalpies of formation of $H_2O(l)$ and $H_2O_2(l)$ are -286 and -188 kJ/mol , respectively. Molar enthalpies of vaporisation of $H_2O(l)$ and $H_2O_2(l)$ are 44 and 53 kJ , respectively. The bond dissociation enthalpy of $O_2(g)$ is 498 kJ/mol . Calculate the bond dissociation enthalpy (in kJ/mol) of O-O bond in H_2O_2 , assuming that the bond dissociation enthalpy of O-H bond is same in both H_2O and H_2O_2 .

- Q.25 The heat evolved on combustion of 1 gm of starch, $(C_6H_{10}O_5)_x$, into $CO_2(g)$ and $H_2O(l)$ at constant pressure, is 4.00 Kcal . Standard Enthalpy of formation of $CO_2(g)$ and $H_2O(l)$ are -94.00 and -65.40 Kcal/mol . The magnitude of standard enthalpy of formation of starch (in Cal/gm) is :

- Q.26 The enthalpy change for the reaction $C_3H_8(g) + H_2(g) \longrightarrow C_2H_6(g) + CH_4(g)$ at 25°C is -55.8 kJ/mol . Calculate the enthalpy of combustion of $C_2H_6(g)$. The enthalpy of combustion of H_2 , & CH_4 are -285.8 & -890.0 kJ/mol respectively. Enthalpy of combustion of propane is $-2220 \text{ kJ mol}^{-1}$.

- Q.27 If the enthalpy of formation of $HCl(g)$ and $Cl^-(aq)$ are -92.3 kJ/mole and -167.3 kJ/mol , find the enthalpy of solution of hydrogen chloride gas.

- Q.28 Benzoic acid is a common standard used in Bomb calorimeters, which maintain a constant volume. If 1.22 gm of benzoic acid gives off 31.7 J of energy when burned in the presence of sufficient oxygen at an initial temperature of 24.6°C , calculate molar heat capacity at constant volume of final product mixture if final temperature is 49.6°C . Also calculate, W and ΔU for the given amount, assuming ideal gas behaviour and heat release in process is only used to rise the temperature.

Q.29 Calculate the electron gain enthalpy of fluorine atom using the following data . Make Born – Haber's cycle. All the values are in kJ mol^{-1} at 25°C .

$$\Delta H_{\text{diss}}(\text{F}_2) = 160, \Delta_f H(\text{NaF(s)}) = -570, \text{I.E.}[\text{Na(g)}] = 495, \Delta H_{\text{Sub}}[\text{Na(s)}] = 100 .$$

Lattice enthalpy of $\text{NaF(s)} = 895$

Q.30 Find the Bond enthalpy (**in kJ/mol**) of one "three centre two electron bond" in B_2H_6 $\{\text{B-H-B} \rightarrow 2\text{B(g)} + \text{H(g)}\}$ from the given data.

$$\Delta H_f^0[\text{BH}_3(\text{g})] = 100 \text{ kJ/mol}$$

$$\Delta H_f^0[\text{B}_2\text{H}_6(\text{g})] = 36 \text{ kJ/mol}$$

$$\Delta H_{\text{atm}}[\text{B(s)}] = 565 \text{ kJ/mol}$$

$$\Delta H_{\text{atm}}[\text{H}_2(\text{g})] = 436 \text{ kJ/mol}$$

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LIST OF IMPORTANT FORMULAS

- (1) **Relation between ΔH and ΔU**

$$\Delta H = \Delta U + (\Delta n_g)RT$$

- (2) **Variation of ΔH_r with temperature**

$$\int_{\Delta H_1}^{\Delta H_2} d(\Delta H) = \int_{T_1}^{T_2} d(\Delta C_p)_r dT$$

$$\frac{\Delta H_2 - \Delta H_1}{T_2 - T_1} = (\Delta C_p)_r$$

If $(C_p)_r$ is constant

- (3) **Calculation of $\Delta H_{\text{reaction}}$ using**

$$(i) \Delta_f H : \Delta_r H = \sum_i a_i \Delta_f H_{(\text{Products})} - \sum_i b_i \Delta_f H_{(\text{reactants})}$$

where a_i and b_i represent the coefficients of the products and reactants in the balanced chemical equation.

$$(ii) \Delta_c H : \Delta_r H = \sum_i b_i \Delta_c H_{(\text{reactants})} - \sum_i a_i \Delta_c H_{(\text{Products})}$$

where a_i and b_i represent the coefficients of the products and reactants in the balanced chemical equation.

- (iii) **Bond Energy**

In gas phase reactions, the standard enthalpy of reaction, $\Delta_r H^\circ$, is related with the bond enthalpies of reactants and products as

$$\Delta_r H^\circ = \sum \text{bond enthalpies (reactants)} - \sum \text{bond enthalpies (products)}$$

- (4) **Relation between I.E. (Ionisation Energy) and $\Delta_i H$ (Ionisation Enthalpy) :**

$$\Delta_i H (\text{Ionisation Enthalpy}) = \text{I.E. (Ionisation Energy)} + \frac{5}{2} RT$$

- (5) **Relation between E.A (Electron Affinity) & $\Delta_{eg} H$ (Electron Gain Enthalpy) :**

$$\Delta_{eg} H = -\text{E.A (Electron Affinity)} - \frac{5}{2} RT$$

- (6) **Resonance Energy :**

$$\text{Resonance Energy} = (\Delta_f H)_{\text{observed}} - (\Delta_f H)_{\text{Calculated}}$$

NOTE: *The reference state of commonly used elements are :*

Elements	Reference state
C	C _(graphite)
S	S _{8(Rhombic)}
P	P _{4(white)}
O	O _{2(g)}
H	H _{2(g)}
Br	Br _{2(l)}
Metal	M _(s) [except Hg _(l)]

ANSWER KEY

Q.1	A	Q.2	C	Q.3	A	Q.4	B	Q.5	A
Q.6	B	Q.7	D	Q.8	B	Q.9	A	Q.10	B
Q.11	C	Q.12	C	Q.13	C	Q.14	C	Q.15	C
Q.16	C	Q.17	C	Q.18	D	Q.19	ACD	Q.20	ABD
Q.21	6	Q.22	0500	Q.23	272	Q.24	142	Q.25	1500
Q.26	−1560	Q.27	−75	Q.28 C = 12.68 J/mole-K, W = 0, ΔU = −31.7 J					
Q.29	−350	Q.30	E _{B-H-B} = 455 kJ/mole						

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